Title

Kernelized polynomial regression

Infinitely many features, and how to afford them

Course Advanced Machine Learning, ETH Zurich Pages 50

Lecturer Carlos Cotrini Date October 5 and 8, 2026



Henri Manuel, before 1912
Portrait of Henri Poincaré

Henri Poincaré

1854 to 1912



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La Science et l'Hypothèse, 1902



Jean Fouquet, c. 1470
The Temple of Jerusalem

Ne pouvons-nous nous contenter de l'expérience toute nue ? Non, cela est impossible ; [. . .] Le savant doit ordonner ; on fait la science avec des faits comme une maison avec des pierres ; mais une accumulation de faits n'est pas plus une science qu'un tas de pierres n'est une maison. Et avant tout le savant doit prévoir.

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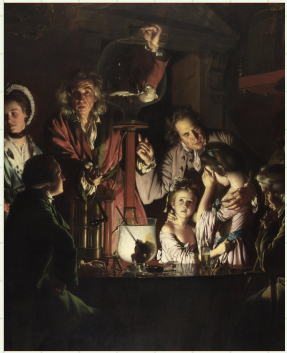


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Can we not be content with bare experience? No, that is impossible; [. . .] The scientist must put things in order; one makes science out of facts as a house out of stones; but an accumulation of facts is no more a science than a heap of stones is a house. And above all the scientist must foresee.

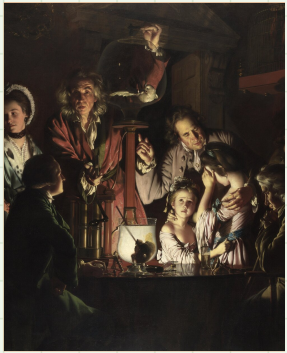
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Joseph Wright of Derby, 1768
Experiment on a Bird in the Air Pump

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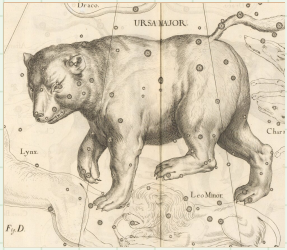


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It is often said that one must experiment without preconceived ideas. That is not possible; it would make all experiment sterile, and one could not do it even if one wished. [. . .] We must, for example, use language, and our language is made of preconceived ideas and can be made of nothing else.

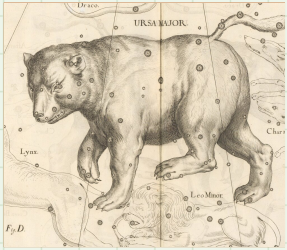
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Il est clair qu'un fait quelconque peut se généraliser d'une infinité de manières, et il s'agit de choisir ; le choix ne peut être guidé que par des considérations de simplicité. [. . .] Nous faisons passer un trait continu, aussi régulier que possible, entre les points donnés par l'observation. [. . .] C'est parce que nous savons d'avance, ou que nous croyons savoir que la loi à exprimer ne peut pas être si compliquée que cela.

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It is clear that any fact can be generalized in infinitely many ways, and one must choose; the choice can only be guided by considerations of simplicity. [. . .] We draw a continuous line, as regular as possible, between the points given by observation. [. . .] It is because we know in advance, or believe we know, that the law to be expressed cannot be as complicated as that.

Henri Poincaré, *La Science et l'Hypothèse* (Paris, 1902), chap. IX.



Leonardo da Vinci, 1498
Ycocedron abscisus vacuus

Les mathématiciens n'étudient pas des objets, mais des relations entre les objets ; il leur est donc indifférent de remplacer ces objets par d'autres, pourvu que les relations ne changent pas. La matière ne leur importe pas, la forme seule les intéresse.

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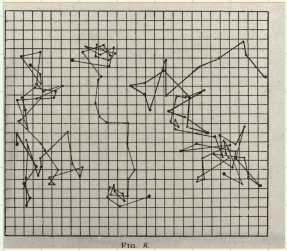


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Mathematicians study relations between objects, so it is indifferent to them to replace these objects by others, provided the relations do not change. Matter is of no importance to them; only form interests them.

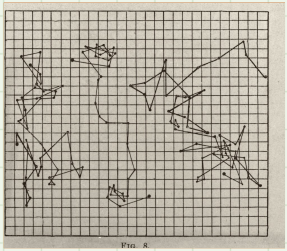
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Tracings of a Brownian grain

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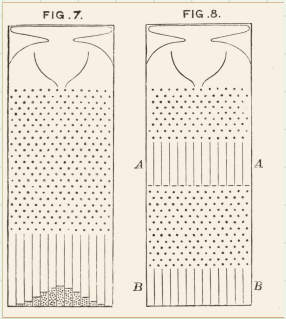


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In the kinetic theory of gases one considers molecules with great speeds, whose trajectories, deformed by incessant collisions, have the most capricious shapes and furrow space in every direction. The observable result is the simple law of Mariotte; each individual fact was complicated; the law of large numbers has restored simplicity in the average.

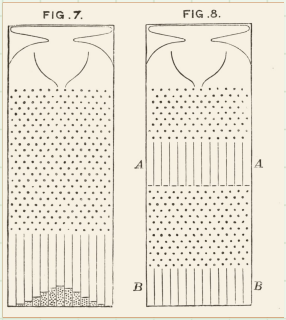
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[...] D'abord, il arrivera souvent que, pour le deviner, ou plutôt pour en deviner ce qui nous est utile, il ne sera pas nécessaire d'en pénétrer le mécanisme ; la loi des grands nombres suffira. [...] Et, en effet, par l'action des moyennes et grâce à la symétrie du milieu, toutes les différences se nivellent et, quelle que soit l'hypothèse faite, le résultat est toujours le même.

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[...] First, it will often happen that, to guess it, or rather to guess what is useful to us about it, it will not be necessary to penetrate its mechanism; the law of large numbers will suffice. [...] And indeed, through the action of averages and thanks to the symmetry of the medium, all differences level out and, whatever the hypothesis made, the result is always the same.

Henri Poincaré, *La Science et l'Hypothèse* (Paris, 1902), chap. IX.



Above all, the scientist must foresee.

From the problem to least squares



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No experiment without ideas; simplicity chooses among generalizations.

Bias, variance and ridge
Polynomial regression



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Only the relations between objects matter.

The kernel trick



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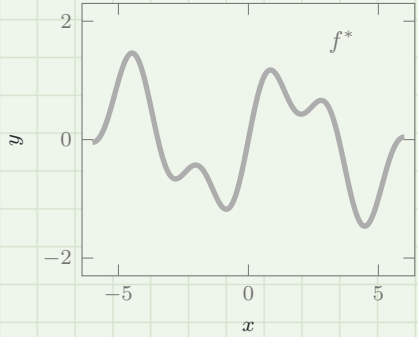
Complicated individuals, a simple average; large numbers suffice.

Random features
Why random features work

Entry 1

From the problem to least squares

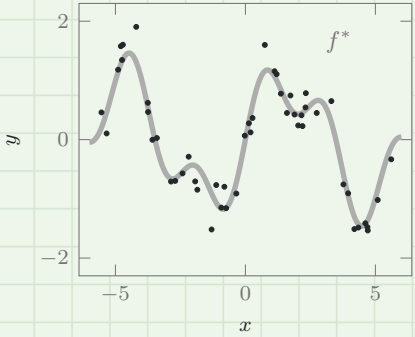
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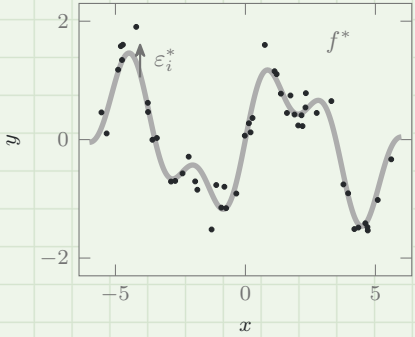


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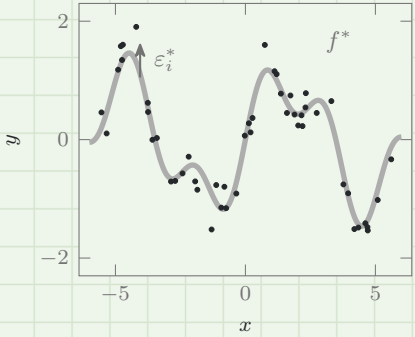
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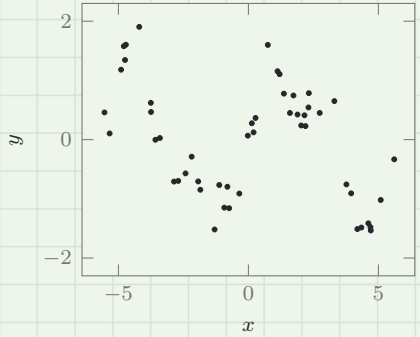
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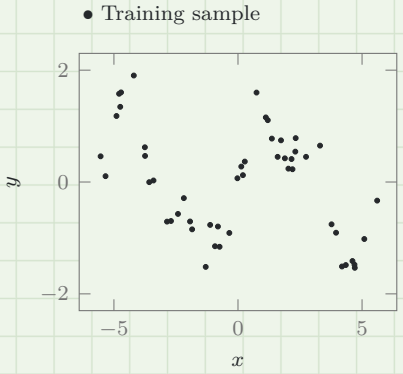
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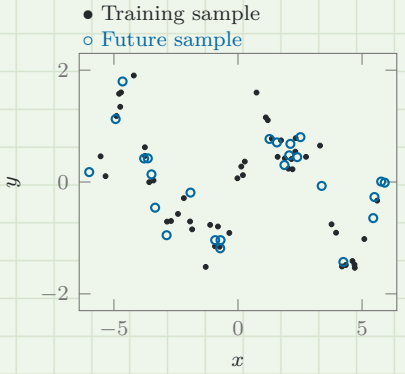
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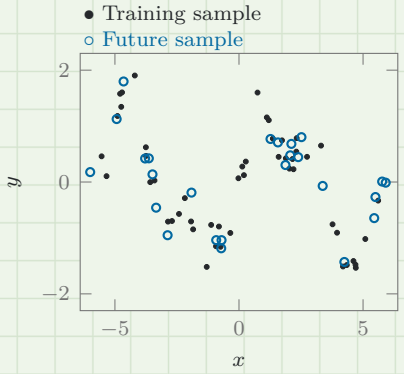


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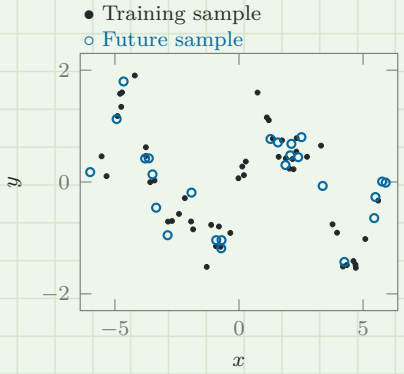
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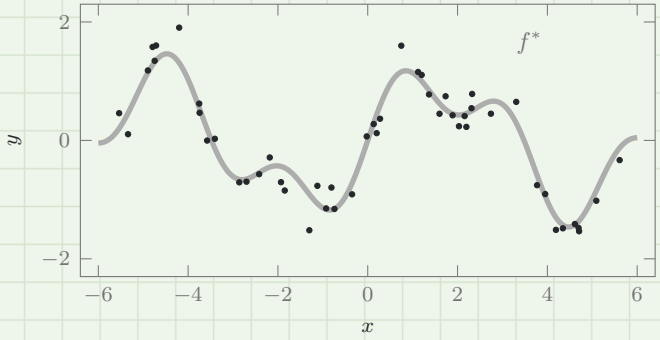
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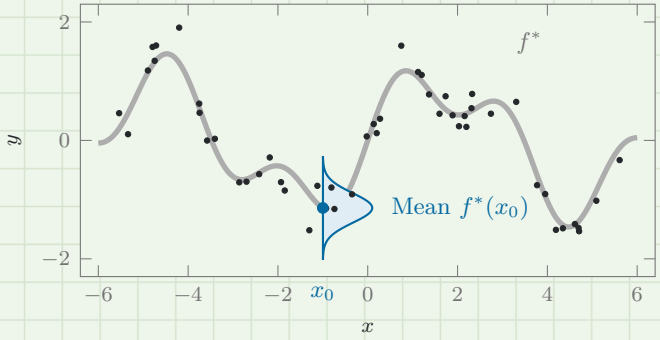
Expected risk



$$R(f) = \underbrace{\mathbb{E}\big[(Y - f(X))^2\big]}_{\text{the error on a new pair } (X,Y)}$$



$$R(f) = \underbrace{\mathbb{E}\big[(f(X) - f^*(X))^2\big]}_{\text{distance to } f^*} + \underbrace{\sigma^2}_{\text{noise}}$$

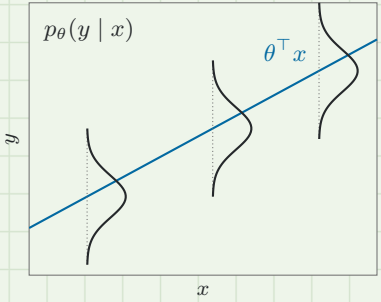


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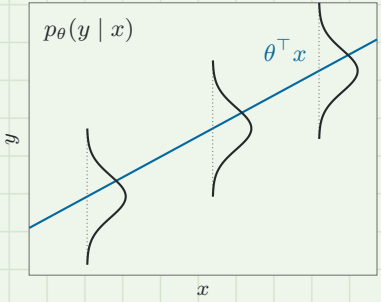
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2) Compute
   $\hat{\theta} = \underset{\theta}{\operatorname{argmax}}\ p_\theta(x_1, y_1, \ldots, x_n, y_n)$



$$p_\theta(x_1, y_1, \ldots, x_n, y_n) \propto \prod_{i \leq n} p_\theta(y_i \mid x_i)$$

Independent pairs, and $p(x_i)$ is free of θ

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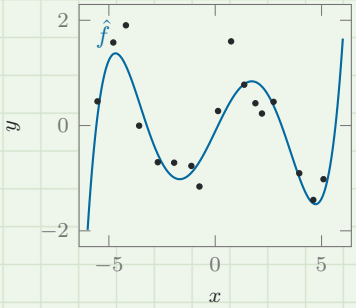
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$$\text{So} \quad \hat{\theta} = \operatorname*{argmin}_{\theta \in \mathbb{R}^d} \sum_{i \le n} (y_i - \theta^\top x_i)^2$$

exp is increasing

1) Approximate $R(\hat{f})$ by

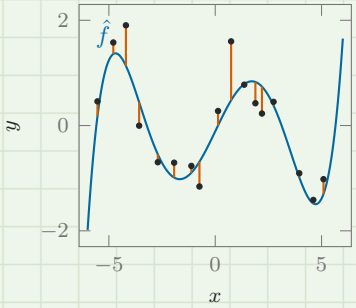
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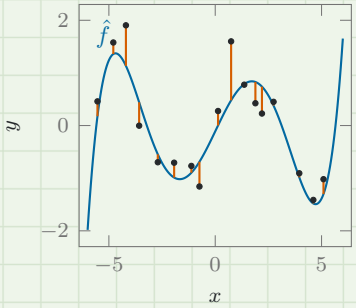
$$\hat{R}(\hat{f}) = \frac{1}{n}\sum_{i\leq n}(y_i - \hat{f}(x_i))^2$$

Empirical risk

2) Choose a parameterized model $\mathcal{H}$, for example

$\mathcal{H}_l = \{f : \mathbb{R}^d \to \mathbb{R} \mid f(x) = w^\top x \text{ for some } w \in \mathbb{R}^d\}$

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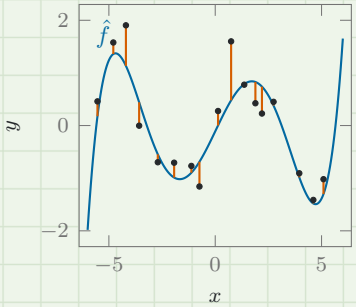
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$$\iff \quad \hat{w} = \underset{w \in \mathbb{R}^d}{\operatorname{argmin}} \sum_{i \leq n} \big(y_i - w^\top x_i\big)^2$$

Searching for $\hat{f}$ is searching for $\hat{w}$

$$\hat{f} = \underset{f \in \mathcal{H}_l}{\operatorname{argmin}} \sum_{i \leq n} \big(y_i - f(x_i)\big)^2$$

$$\iff \quad \hat{w} = \underset{w \in \mathbb{R}^d}{\operatorname{argmin}} \sum_{i \leq n} \big(y_i - w^\top x_i\big)^2 \qquad \text{Searching for } \hat{f} \text{ is searching for } \hat{w}$$

$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y$$

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$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y \longleftarrow \text{OLSE}$$

$$\hat{f} = \underset{f \in \mathcal{H}_l}{\operatorname{argmin}} \sum_{i \leq n} \big(y_i - f(x_i)\big)^2$$

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$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y \longleftarrow \text{OLSE}$$

$$\text{where } \mathbf{X} = \begin{bmatrix} — \ x_1^\top \ — \\ \vdots \\ — \ x_n^\top \ — \end{bmatrix} \in \mathbb{R}^{n \times d} \quad \text{and} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n$$

Maximum likelihood

$$y_i \sim \mathcal{N}(\theta^\top x_i, \sigma^2)$$

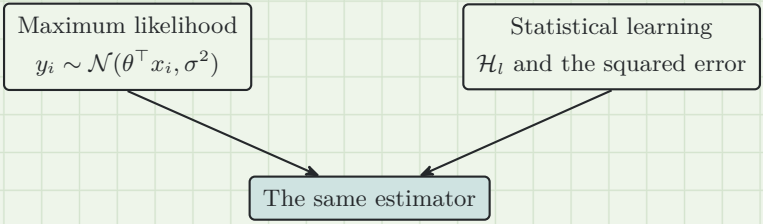
Maximum likelihood

$y_i \sim \mathcal{N}(\theta^\top x_i, \sigma^2)$

Statistical learning

$\mathcal{H}_l$ and the squared error

$$\hat{\theta}_{\mathrm{MLE}} = \operatorname*{argmin}_{\theta \in \mathbb{R}^d} \sum_{i \le n} (y_i - \theta^\top x_i)^2 = \hat{w}_{\mathrm{OLSE}}$$



Build $\mathbf{X}^\top\mathbf{X}$: d^2 inner products $O(nd^2)$



$n \times d$

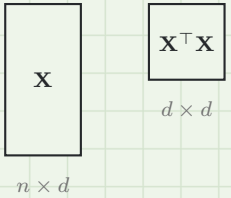
Build $\mathbf{X}^\top \mathbf{X}$: d^2 inner products $O(nd^2)$

Build $\mathbf{X}^\top y$: d inner products $O(nd)$

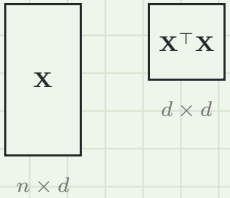


$n \times d$

Build $\mathbf{X}^\top\mathbf{X}$: d^2 inner products	$O(nd^2)$
Build $\mathbf{X}^\top y$: d inner products	$O(nd)$
Solve the $d \times d$ system	$O(d^3)$

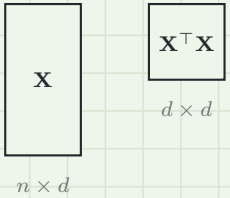


Build $\mathbf{X}^\top\mathbf{X}$: d^2 inner products	$O(nd^2)$
Build $\mathbf{X}^\top y$: d inner products	$O(nd)$
Solve the $d \times d$ system	$O(d^3)$
Total	$O(nd^2 + d^3)$



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Solve the $d \times d$ system	$O(d^3)$
Total	$O(nd^2 + d^3)$

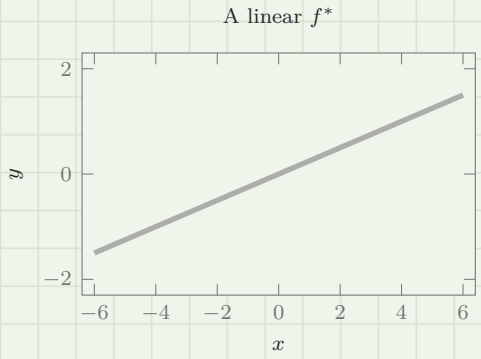
Each inner product runs over the n points: linear in the data, cubic in d



Entry 2

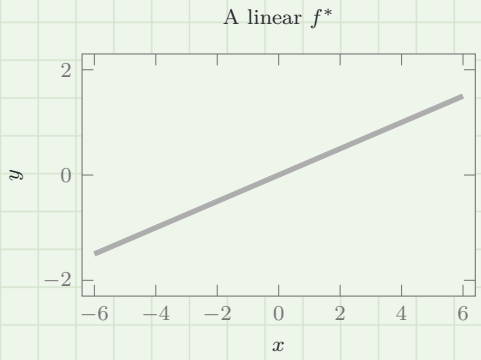
Bias, variance and ridge

If $y_i = x_i^\top w^* + \varepsilon_i$ for some w^*, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, then $\mathbb{E}_{y|\mathbf{X}}[\hat{w}] = \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y]$



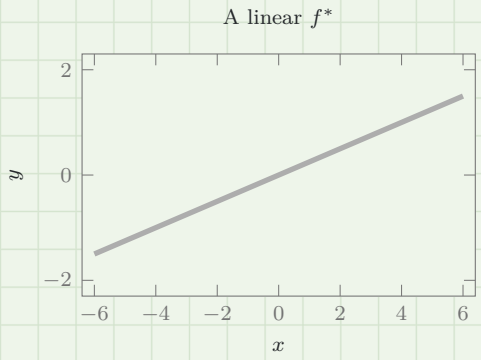
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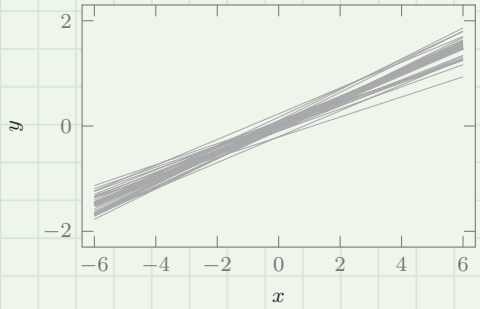
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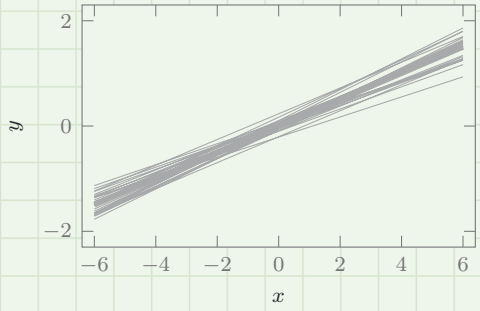
A linear f^*



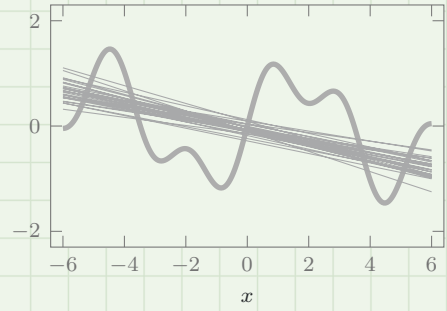
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A linear f^*



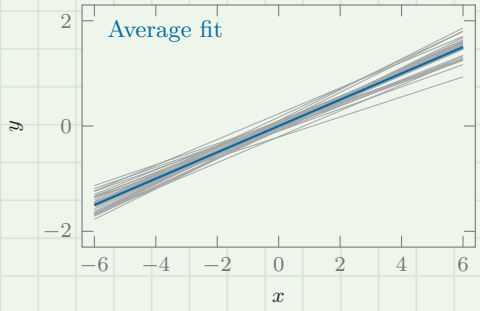
Our f^*



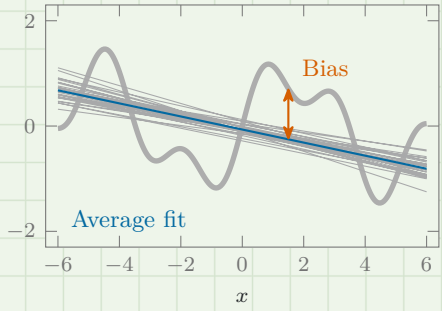
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Our f^*



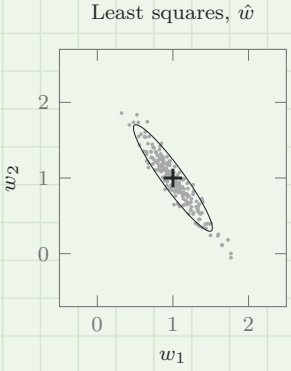
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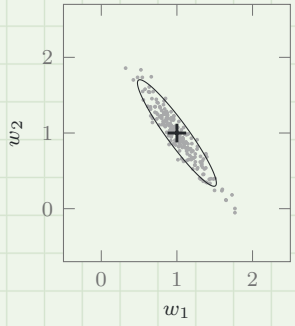
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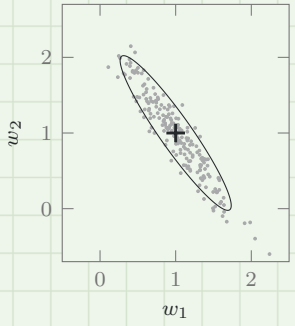
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Least squares, $\hat{w}$



Least squares on half the data, a $\tilde{w}$



$$\text{Error} = \text{bias}^2 + \text{variance} + \text{noise}$$

$$\mathbb{E}[(y_* - \hat{f}(x_*))^2] = \underbrace{\left(\mathbb{E}[\hat{f}(x_*)] - f^*(x_*)\right)^2}_{\text{bias}^2} + \underbrace{\text{Var}[\hat{f}(x_*)]}_{\text{variance}} + \underbrace{\sigma^2}_{\text{noise}}$$

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- OLSE usually has no bias, but a lot of variance

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- OLSE usually has no bias, but a lot of variance
- Adding a bit of bias may lead to a substantial loss of variance

Ridge estimator

$$\hat{w}_r = \underset{w}{\operatorname{argmin}} \ \hat{R}_r(w)$$

Ridge estimator

$$\hat{w}_r = \operatorname*{argmin}_w \hat{R}_r(w)$$

$$\hat{R}_r(w) = \underbrace{\|y - \mathbf{X}w\|^2}_{\text{fit the data}} + \underbrace{\lambda \|w\|^2}_{\text{keep } w \text{ small}}$$

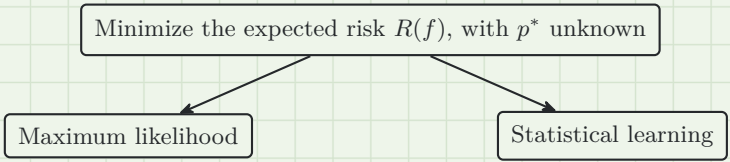
Ridge estimator

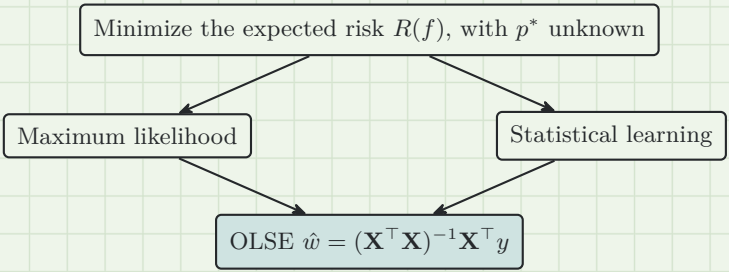
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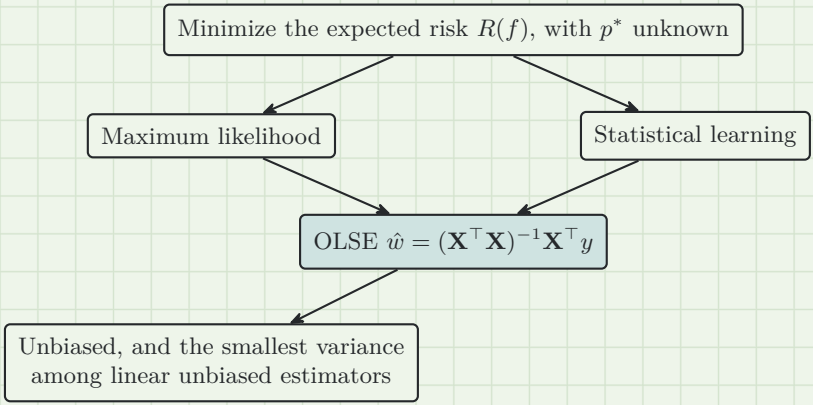
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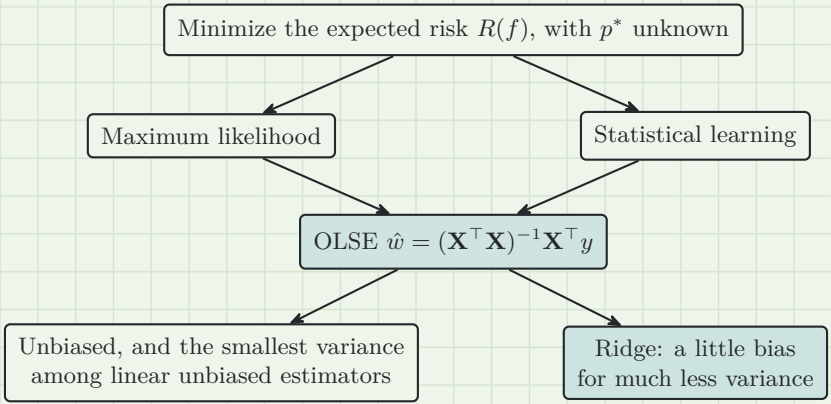
$$\hat{w}_r = (\mathbf{X}^\top \mathbf{X} + \lambda I)^{-1} \mathbf{X}^\top y$$

Minimize the expected risk $R(f)$, with p^* unknown


Minimize the expected risk $R(f)$, with p^* unknown
Maximum likelihood
Statistical learning

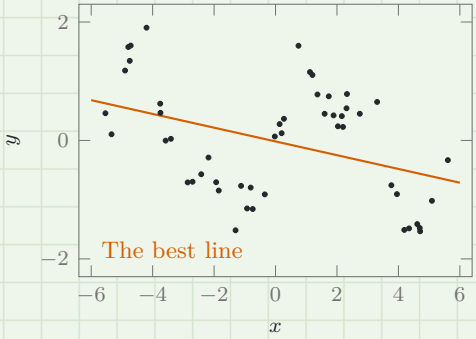

Minimize the expected risk $R(f)$, with p^* unknown
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Statistical learning
OLSE $\hat{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top y$

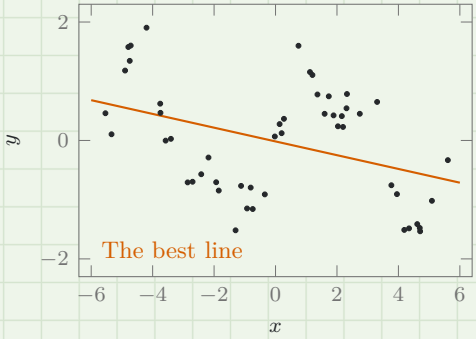

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Unbiased, and the smallest variance among linear unbiased estimators


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Unbiased, and the smallest variance among linear unbiased estimators
Ridge: a little bias for much less variance

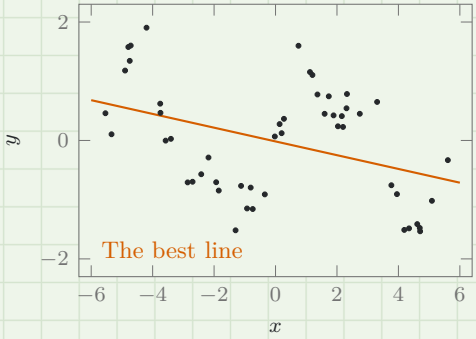
Entry 3

Polynomial regression


2
0
−2
y
The best line
−6
−4
−2
0
2
4
6
x



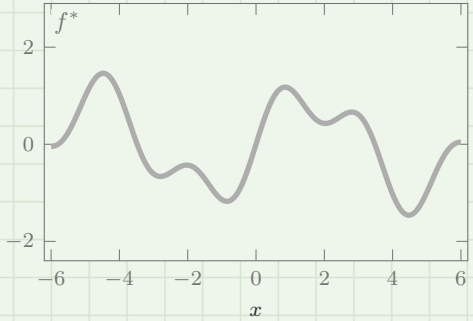
A universal approximator would be better

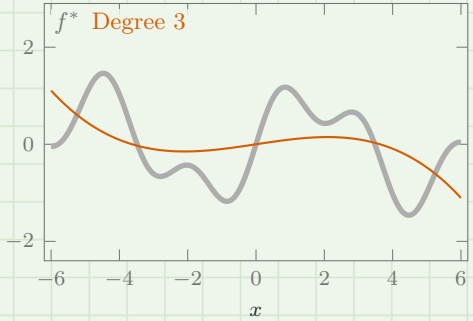


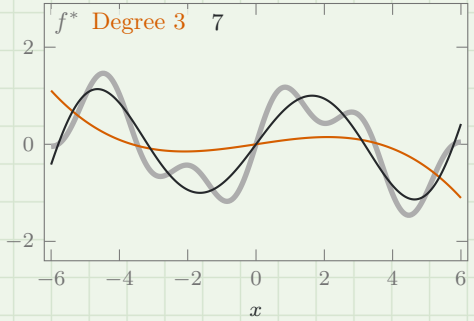
A universal approximator would be better

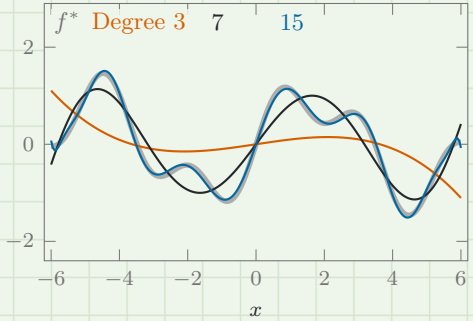
Universal approximator

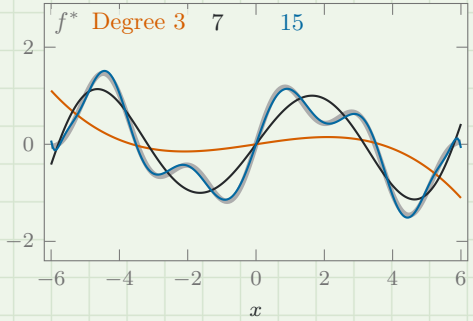
$\mathcal{H}$ is a universal approximator if, for every closed and bounded $B \subset \mathbb{R}^d$, every continuous $g : B \to \mathbb{R}$ and every $\delta > 0$, there is an $f \in \mathcal{H}$ with $|f(x) - g(x)| < \delta$ for all $x \in B$


f^*
2
0
−2
−6
−4
−2
0
2
4
6
x


f^*
Degree 3
2
0
−2
−6
−4
−2
0
2
4
6
x

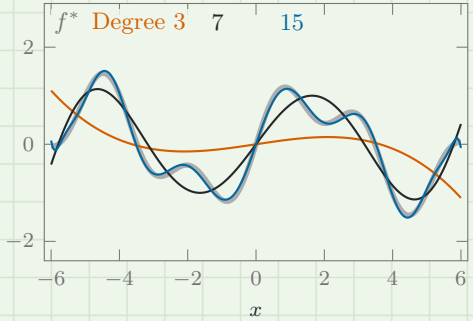

f^*
Degree 3
7
2
0
−2
−6
−4
−2
0
2
4
6
x


f^*
Degree 3
7
15
2
0
−2
−6
−4
−2
0
2
4
6
x



- Weierstrass: on a closed bounded domain, polynomials come as close to any continuous function as we like

Weierstrass (1885), generalized by Stone (1948)



- Weierstrass: on a closed bounded domain, polynomials come as close to any continuous function as we like
- Raise the degree, lower the error

Weierstrass (1885), generalized by Stone (1948)

How do we compute $\hat{f} = \underset{f \in \mathcal{H}_p}{\operatorname{argmin}} \ \hat{R}(f)$?

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We note that there is $\tilde{\varphi} : \mathbb{R}^d \to \mathbb{R}^\infty$ such that, for every $f \in \mathcal{H}_p$, there is a $w_f \in \mathbb{R}^\infty$ with

$$f(x) = w_f^\top \tilde{\varphi}(x)$$

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⚠ Infinitely many weights

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Infinitely many weights

$\mathcal{H}_l$

$x \longrightarrow w^\top x$

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⚠ Infinitely many weights

$\mathcal{H}_l$ $\quad x \longrightarrow w^\top x$

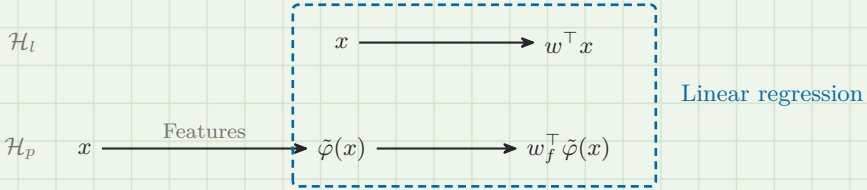
$\mathcal{H}_p$ $\quad x \xrightarrow{\text{Features}} \tilde{\varphi}(x) \longrightarrow w_f^\top \tilde{\varphi}(x)$

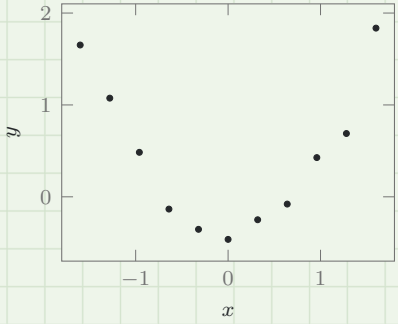
How do we compute $\hat{f} = \underset{f \in \mathcal{H}_p}{\operatorname{argmin}} \ \hat{R}(f)$?

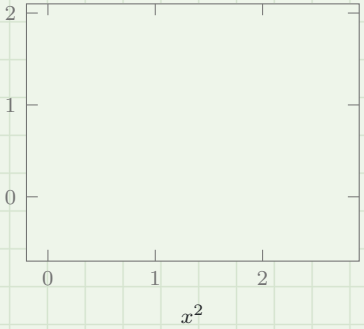
We note that there is $\tilde{\varphi} : \mathbb{R}^d \to \mathbb{R}^\infty$ such that, for every $f \in \mathcal{H}_p$, there is a $w_f \in \mathbb{R}^\infty$ with

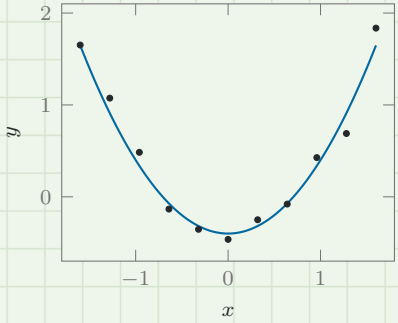
$$f(x) = w_f^\top \tilde{\varphi}(x)$$

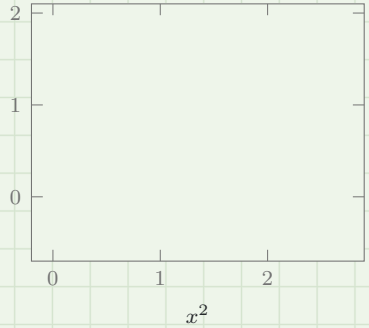
Infinitely many weights

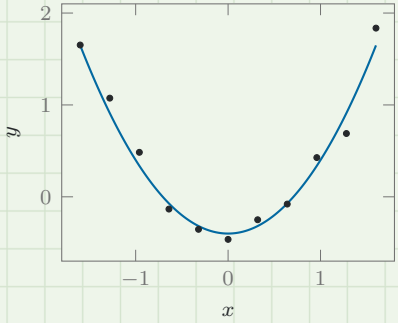


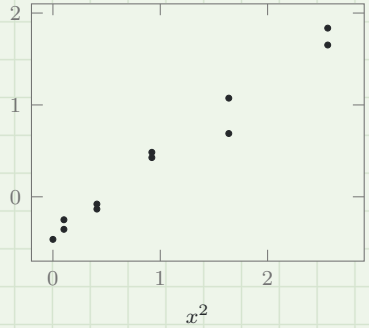

2
1
0
y
−1
0
1
x

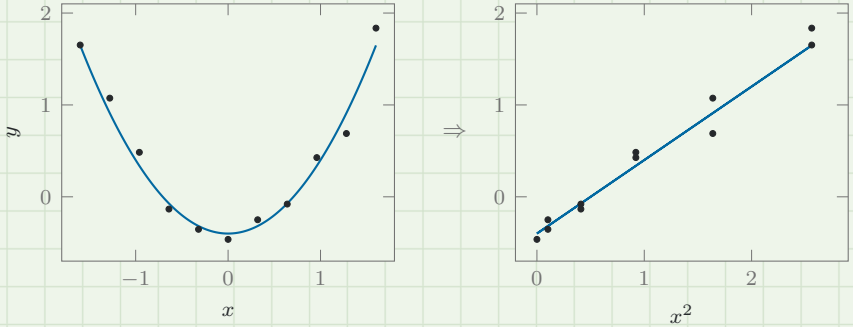

2
1
0
0
1
2
x^2

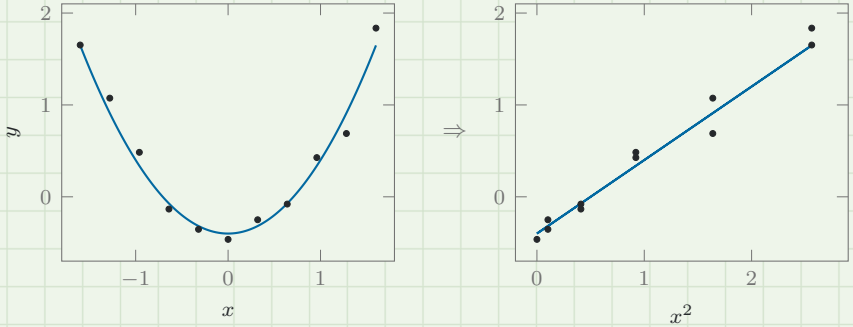

2
1
0
y
−1
0
1
x


2
1
0
0
1
2
x^2

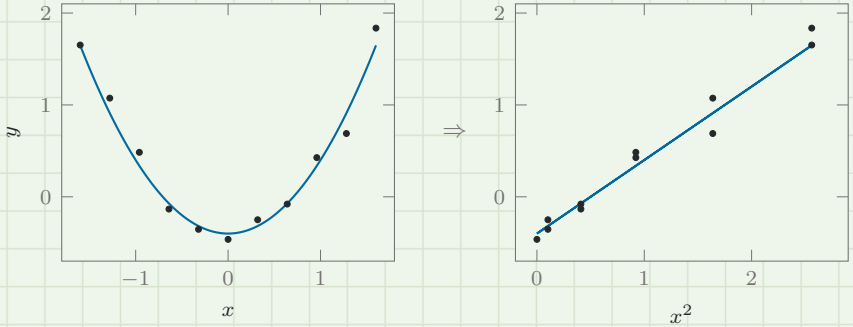

2
1
0
−1
0
1
y
x


2
1
0
0
1
2
x^2


2
1
0
y
−1
0
1
x
⇒
2
1
0
0
1
2
x^2



- Same data, new coordinate



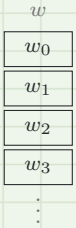
- Same data, new coordinate
- Polynomial regression is linear regression on features

$$f(x) = w_0 + w_1 x + w_2 x^2 + w_3 x^3 + \cdots$$

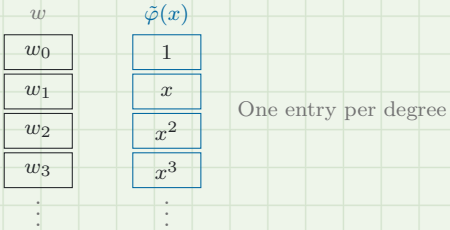
$$f(x) = \sum_{k \in \mathbb{N}} w_k x^k$$

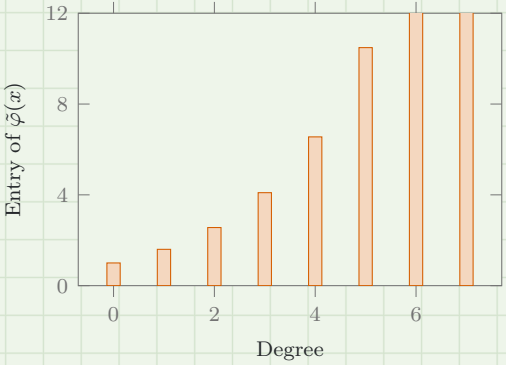
$$f(x) = \sum_{k \in \mathbb{N}} w_k x^k = \langle w, \tilde{\varphi}(x) \rangle$$

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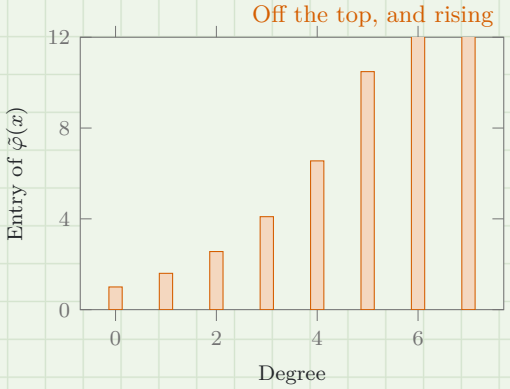


$$f(x) = \sum_{k \in \mathbb{N}} w_k x^k = \langle w, \tilde{\varphi}(x) \rangle$$

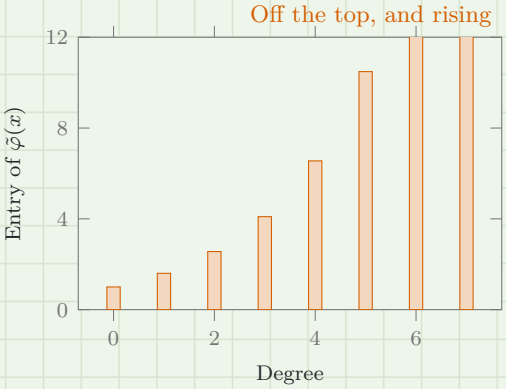




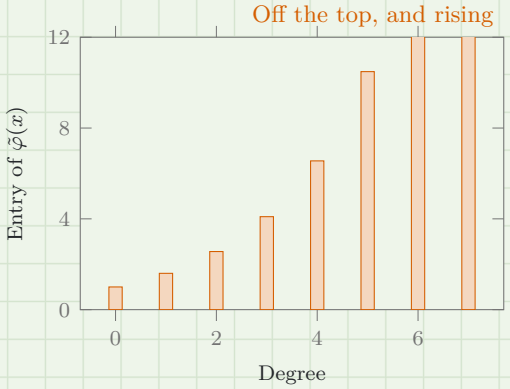
$$\langle w, \tilde{\varphi}(x) \rangle = \sum_{k \in \mathbb{N}} w_k x^k$$



$$\langle w, \tilde{\varphi}(x) \rangle = \sum_{k \in \mathbb{N}} w_k x^k$$

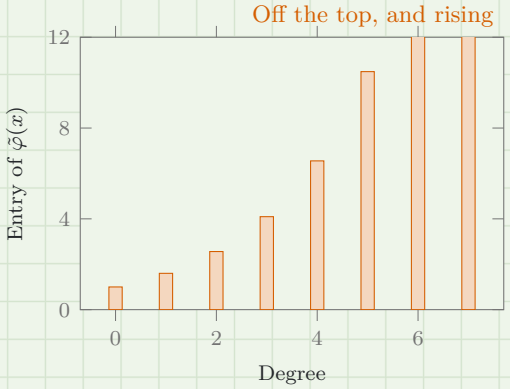


$$\langle w, \tilde{\varphi}(x) \rangle = \underbrace{1 + 1 + 1 + \cdots}_{x=1,\ w=(1,1,1,\ldots)} = \infty$$



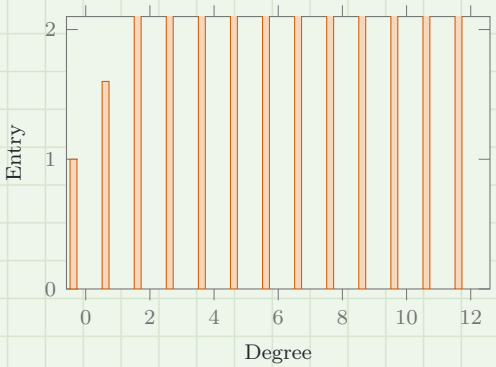
$$\langle w, \tilde{\varphi}(x) \rangle = \underbrace{1 + 1 + 1 + \cdots}_{x=1,\ w=(1,1,1,\ldots)} = \infty$$

- High degrees are too large

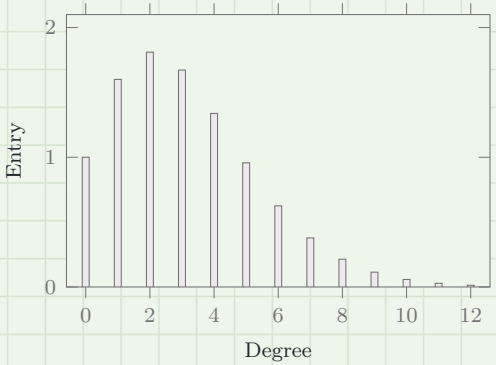


$$\langle w, \tilde{\varphi}(x) \rangle = \underbrace{1 + 1 + 1 + \cdots}_{x=1,\ w=(1,1,1,\ldots)} = \infty$$

- High degrees are too large
- Linear regression is built from inner products

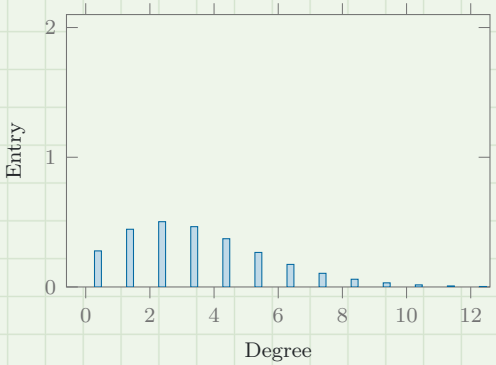


x^k



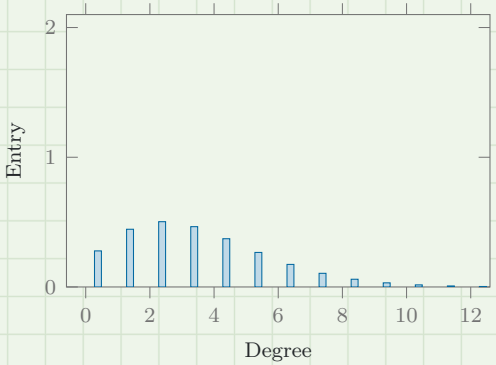
$$\frac{x^k}{\sqrt{k!}}$$

- Factorials beat powers



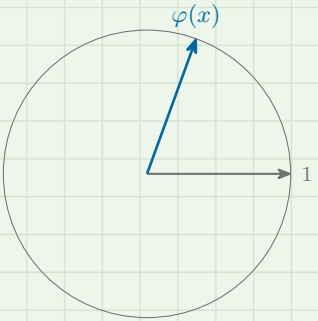
$$\varphi(x) = e^{-x^2/2}\left[\frac{x^k}{\sqrt{k!}}\right]_{k\in\mathbb{N}}$$

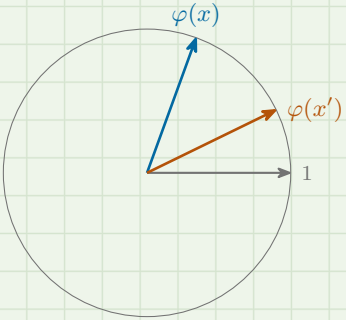
- Factorials beat powers
- The bump tames what is left

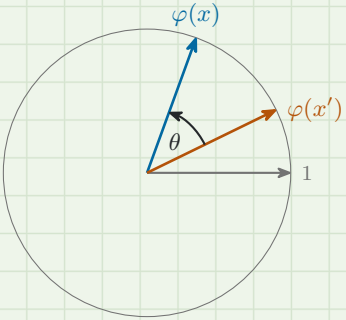


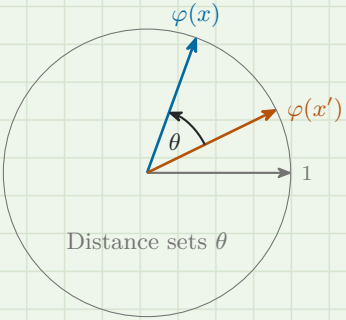
$$\varphi(x) = e^{-x^2/2}\left[\frac{x^k}{\sqrt{k!}}\right]_{k\in\mathbb{N}}$$

- Factorials beat powers
- The bump tames what is left
- Every $\|\varphi(x)\| = 1$

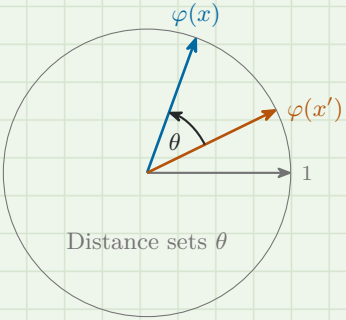

$\varphi(x)$
1


$\varphi(x)$
$\varphi(x')$
1


$\varphi(x)$
$\varphi(x')$
θ
1


$\varphi(x)$
$\varphi(x')$
θ
1
Distance sets θ

$$\langle \varphi(x), \varphi(x') \rangle = \cos\theta$$



$\alpha_2 =$	0	1	2	$\cdots$
$\alpha_1 = 0$	w_{00}	$w_{01}x_2$	$w_{02}x_2^2$	$\cdots$
1				
2				
$\vdots$				

$f(x_1, x_2)$ is the sum of every cell

$\alpha_2 =$	0	1	2	$\cdots$
$\alpha_1 = 0$	w_{00}	$w_{01}x_2$	$w_{02}x_2^2$	$\cdots$
1	$w_{10}x_1$	$w_{11}x_1x_2$	$w_{12}x_1x_2^2$	$\cdots$
2				
$\vdots$				

$f(x_1, x_2)$ is the sum of every cell

$\alpha_2 =$	0	1	2	$\cdots$
$\alpha_1 = 0$	w_{00}	$w_{01}x_2$	$w_{02}x_2^2$	$\cdots$
1	$w_{10}x_1$	$w_{11}x_1x_2$	$w_{12}x_1x_2^2$	$\cdots$
2	$w_{20}x_1^2$	$w_{21}x_1^2x_2$	$w_{22}x_1^2x_2^2$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

$f(x_1, x_2)$ is the sum of every cell

$\alpha_2 =$	0	1	2	$\cdots$
$\alpha_1 = 0$	w_{00}	$w_{01}x_2$	$w_{02}x_2^2$	$\cdots$
1	$w_{10}x_1$	$w_{11}x_1x_2$	$w_{12}x_1x_2^2$	$\cdots$
2	$w_{20}x_1^2$	$w_{21}x_1^2x_2$	$w_{22}x_1^2x_2^2$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

$$f(x_1, x_2) = \sum_{\alpha_1 \in \mathbb{N}} \sum_{\alpha_2 \in \mathbb{N}} w_{\alpha_1 \alpha_2} \, x_1^{\alpha_1} x_2^{\alpha_2}$$

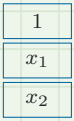
$\alpha_2 =$	0	1	2	$\cdots$
$\alpha_1 = 0$	w_{00}	$w_{01}x_2$	$w_{02}x_2^2$	$\cdots$
1	$w_{10}x_1$	$w_{11}x_1x_2$	$w_{12}x_1x_2^2$	$\cdots$
2	$w_{20}x_1^2$	$w_{21}x_1^2x_2$	$w_{22}x_1^2x_2^2$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

$$f(x_1, x_2) = \sum_{\alpha=(\alpha_1,\alpha_2)\in\mathbb{N}^2} w_\alpha \, x^\alpha = \langle w, \tilde{\varphi}(x) \rangle$$

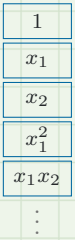
$$f(x) = f(x_1, \ldots, x_d) = \sum_{\alpha \in \mathbb{N}^d} w_\alpha \, x^\alpha = \langle w, \tilde{\varphi}(x) \rangle, \qquad x^\alpha = \prod_{i \leq d} x_i^{\alpha_i}$$

With the correction we fit $f(x) = \langle w, \varphi(x) \rangle$, $\varphi(x) = e^{-\|x\|^2/2} \left[\frac{x^\alpha}{\sqrt{\alpha!}} \right]_{\alpha \in \mathbb{N}^d}$

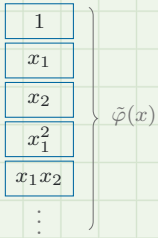
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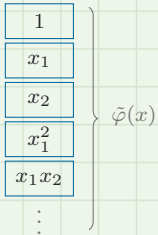
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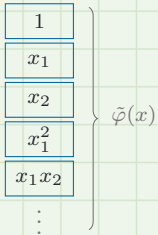


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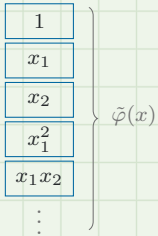
- $\alpha! = \prod_{i \le d} \alpha_i!$, one factorial per variable

With the correction we fit $f(x) = \langle w, \varphi(x) \rangle$, $\varphi(x) = e^{-\|x\|^2/2} \left[\frac{x^\alpha}{\sqrt{\alpha!}} \right]_{\alpha \in \mathbb{N}^d}$



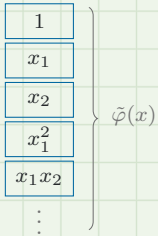
- $\alpha! = \prod_{i \le d} \alpha_i!$, one factorial per variable
- $d = 100$, degree 5: almost 10^8 features

With the correction we fit $f(x) = \langle w, \varphi(x) \rangle$, $\varphi(x) = e^{-\|x\|^2/2} \left[\frac{x^\alpha}{\sqrt{\alpha!}} \right]_{\alpha \in \mathbb{N}^d}$



- $\alpha! = \prod_{i \leq d} \alpha_i!$, one factorial per variable
- $d = 100$, degree 5: almost 10^8 features
- No maximal degree: infinitely many

With the correction we fit $f(x) = \langle w, \varphi(x) \rangle$, $\varphi(x) = e^{-\|x\|^2/2} \left[\frac{x^\alpha}{\sqrt{\alpha!}} \right]_{\alpha \in \mathbb{N}^d}$



- $\alpha! = \prod_{i \leq d} \alpha_i!$, one factorial per variable
- $d = 100$, degree 5: almost 10^8 features
- No maximal degree: infinitely many
- Still a universal approximator

Entry 4

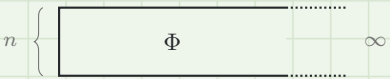
The kernel trick

$$\hat{w}_\infty = \underset{w \in \mathbb{R}^\infty}{\operatorname{argmin}} \|\Phi w - y\|^2$$

with $\Phi = \begin{bmatrix} — \varphi(x_1)^\top — \\ \vdots \\ — \varphi(x_n)^\top — \end{bmatrix}$



$$\hat{w}_\infty = \underset{w \in \mathbb{R}^\infty}{\operatorname{argmin}} \, \|\Phi w - y\|^2$$



$$\text{with } \Phi = \begin{bmatrix} — \; \varphi(x_1)^\top \; — \\ \vdots \\ — \; \varphi(x_n)^\top \; — \end{bmatrix}$$

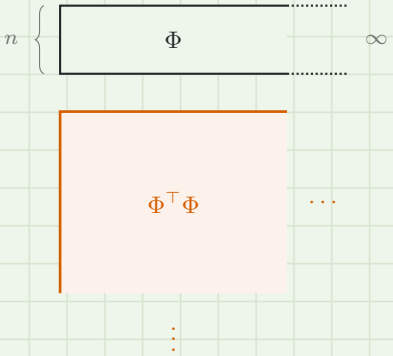
$$\hat{w}_\infty = (\Phi^\top \Phi)^{-1} \Phi^\top y$$

$$\hat{w}_\infty = \underset{w \in \mathbb{R}^\infty}{\operatorname{argmin}} \|\Phi w - y\|^2$$

with $\Phi = \begin{bmatrix} — \varphi(x_1)^\top — \\ \vdots \\ — \varphi(x_n)^\top — \end{bmatrix}$

$$\hat{w}_\infty = (\Phi^\top \Phi)^{-1} \Phi^\top y$$

But $\Phi \in \mathbb{R}^{n \times \infty}$, so $\Phi^\top \Phi \in \mathbb{R}^{\infty \times \infty}$

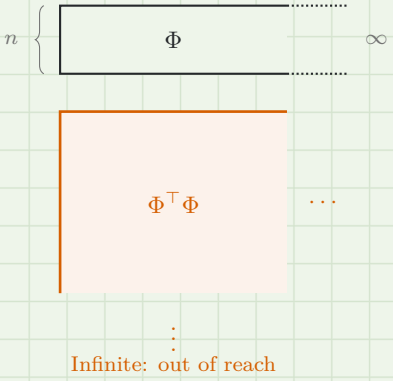


$$\hat{w}_\infty = \underset{w \in \mathbb{R}^\infty}{\operatorname{argmin}} \|\Phi w - y\|^2$$

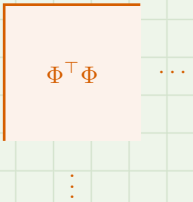
with $\Phi = \begin{bmatrix} — \varphi(x_1)^\top — \\ \vdots \\ — \varphi(x_n)^\top — \end{bmatrix}$

$$\hat{w}_\infty = (\Phi^\top \Phi)^{-1} \Phi^\top y$$

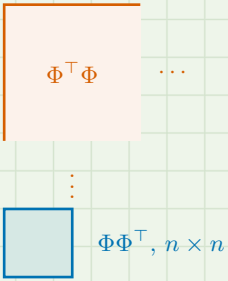
But $\Phi \in \mathbb{R}^{n \times \infty}$, so $\Phi^\top \Phi \in \mathbb{R}^{\infty \times \infty}$



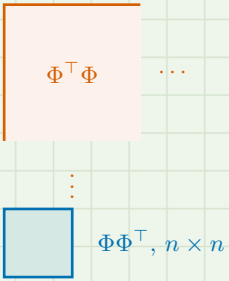
$$\hat{w}_\infty = (\Phi^\top \Phi)^{-1} \Phi^\top y$$



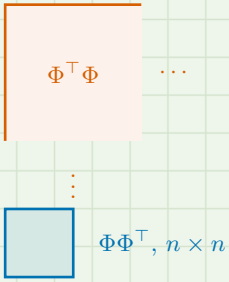
$$\begin{aligned}\hat{w}_\infty &= (\Phi^\top \Phi)^{-1} \Phi^\top y \\ &= (\Phi^\top \Phi)^{-1} \Phi^\top \underbrace{(\Phi\Phi^\top)(\Phi\Phi^\top)^{-1}}_{I_n} y\end{aligned}$$



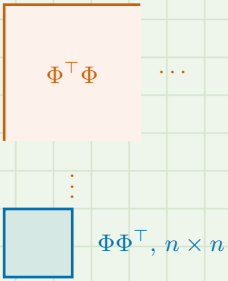
$$\begin{aligned}\hat{w}_\infty &= (\Phi^\top\Phi)^{-1}\Phi^\top y \\ &= (\Phi^\top\Phi)^{-1}\Phi^\top \underbrace{(\Phi\Phi^\top)(\Phi\Phi^\top)^{-1}}_{I_n} y \\ &= \underbrace{(\Phi^\top\Phi)^{-1}(\Phi^\top\Phi)}_{I}\ \Phi^\top(\Phi\Phi^\top)^{-1}y\end{aligned}$$



$$
\begin{aligned}
\hat{w}_\infty &= (\Phi^\top \Phi)^{-1} \Phi^\top y \\
&= (\Phi^\top \Phi)^{-1} \Phi^\top \underbrace{(\Phi\Phi^\top)(\Phi\Phi^\top)^{-1}}_{I_n} y \\
&= \underbrace{(\Phi^\top \Phi)^{-1}(\Phi^\top \Phi)}_{I}\, \Phi^\top (\Phi\Phi^\top)^{-1} y \\
&= \Phi^\top (\Phi\Phi^\top)^{-1} y
\end{aligned}
$$



$$
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\hat{w}_\infty &= (\Phi^\top \Phi)^{-1} \Phi^\top y \\
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&= \Phi^\top (\Phi\Phi^\top)^{-1} y
\end{aligned}
$$

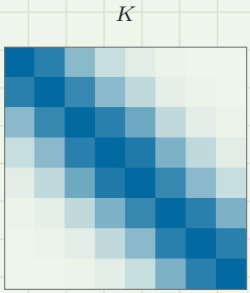


Formally, $\Phi^\top \Phi$ has no inverse: read the middle lines as shorthand. The last line is exact, the least-squares solution of smallest norm

For a test point $x_* \in \mathbb{R}^d$, $\quad \hat{f}(x_*) = \varphi(x_*)^\top \hat{w}_\infty = \varphi(x_*)^\top \Phi^\top (\Phi\Phi^\top)^{-1} y$

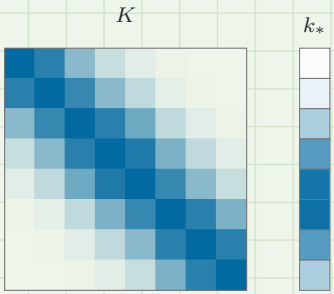
For a test point $x_* \in \mathbb{R}^d$, $\hat{f}(x_*) = \underbrace{\varphi(x_*)^\top \Phi^\top}_{k_*^\top} \big(\underbrace{\Phi\Phi^\top}_{K}\big)^{-1} y$

For a test point $x_* \in \mathbb{R}^d$, $\hat{f}(x_*) = \underbrace{\varphi(x_*)^\top \Phi^\top}_{k_*^\top} \big(\underbrace{\Phi\Phi^\top}_{K}\big)^{-1} y$



$K \in \mathbb{R}^{n \times n}$,
$K_{ij} = \varphi(x_i)^\top \varphi(x_j)$
$= k(x_i, x_j) = \exp(-\frac{1}{2}\|x_i - x_j\|^2)$

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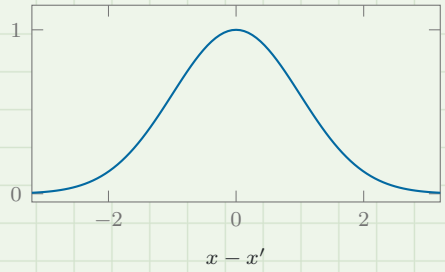
$k_* \in \mathbb{R}^n$, $k_{*i} = \varphi(x_*)^\top \varphi(x_i)$
$= k(x_*, x_i) = \exp\big(-\frac{1}{2}\|x_* - x_i\|^2\big)$

$$\langle \varphi(x), \varphi(x') \rangle = e^{-\frac{\|x\|^2 + \|x'\|^2}{2}} \sum_{\alpha \in \mathbb{N}^d} \prod_{i \leq d} \frac{(x_i x_i')^{\alpha_i}}{\alpha_i!}$$

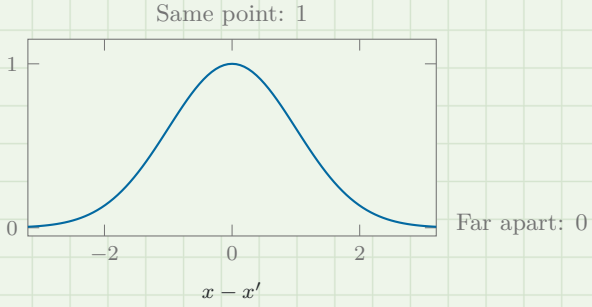
$$\langle \varphi(x), \varphi(x') \rangle = e^{-\frac{\|x\|^2 + \|x'\|^2}{2}} \underbrace{\prod_{i \leq d} e^{x_i x'_i}}_{\text{exponential series}}$$

$$\langle \varphi(x), \varphi(x') \rangle = \underbrace{e^{-\|x-x'\|^2/2}}_{\text{the Gaussian kernel } k(x,x')}$$

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Polynomial OLSE

1) Compute K from $\{x_i\}_{i \leq n}$

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Polynomial OLSE

1) Compute K from $\{x_i\}_{i \leq n}$
2) Compute K^{-1}
3) Given $x_* \in \mathbb{R}^d$, compute k_* from x_* and $\{x_i\}_{i \leq n}$
4) Output $k_*^\top K^{-1} y$

One can show that

$$\hat{w}_\infty = \underset{w \in \mathbb{R}^\infty}{\operatorname{argmin}} \, \|\Phi w - y\|^2 + \lambda \|w\|^2$$

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$$\hat{w}_\infty = \underset{w \in \mathbb{R}^\infty}{\operatorname{argmin}} \|\Phi w - y\|^2 + \lambda \|w\|^2 = \Phi^\top (\Phi \Phi^\top + \lambda I)^{-1} y$$

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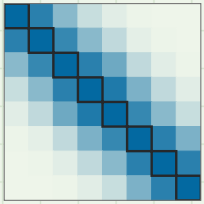
$$\text{so } \hat{f}(x_*) = k_*^\top (K + \lambda I)^{-1} y$$

One can show that

$$\hat{w}_\infty = \operatorname*{argmin}_{w \in \mathbb{R}^\infty} \|\Phi w - y\|^2 + \lambda \|w\|^2 = \Phi^\top (\Phi\Phi^\top + \lambda I)^{-1} y$$

$$\text{so } \hat{f}(x_*) = k_*^\top (K + \lambda I)^{-1} y$$

$K + \lambda I$



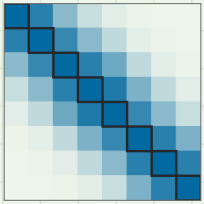
$+\lambda$ on every diagonal entry

One can show that

$$\hat{w}_\infty = \operatorname*{argmin}_{w \in \mathbb{R}^\infty} \|\Phi w - y\|^2 + \lambda \|w\|^2 = \Phi^\top (\Phi\Phi^\top + \lambda I)^{-1} y$$

$$\text{so } \hat{f}(x_*) = k_*^\top (K + \lambda I)^{-1} y$$

$K + \lambda I$



$+\lambda$ on every diagonal entry

Step 2 of the algorithm now inverts $K + \lambda I$

1. Build K $O(n^2 d)$

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2. Solve $(K + \lambda I_n)\beta = y$ $O(n^3)$

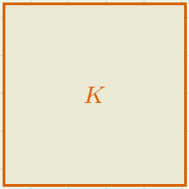
1. Build K $O(n^2 d)$
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1. Build K	$O(n^2 d)$
2. Solve $(K + \lambda I_n)\beta = y$	$O(n^3)$
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	$O(n^2 d + n^3)$

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2. Solve $(K + \lambda I_n)\beta = y$ $O(n^3)$
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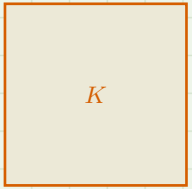
$O(n^2 d + n^3)$



$n = 10^6$

1. Build K — $O(n^2 d)$
2. Solve $(K + \lambda I_n)\beta = y$ — $O(n^3)$
3. Build k_* — $O(nd)$
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$O(n^2 d + n^3)$



$n = 10^6$

- 8 TB of memory

1. Build K — $O(n^2 d)$
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$O(n^2 d + n^3)$



$n = 10^6$

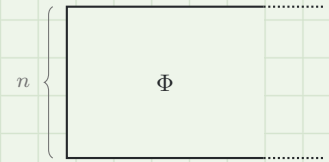
- 8 TB of memory
- 10^{18} operations

Entry 5

Random features

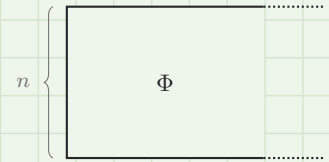
We show that

$$\varphi(x_*)^\top \hat{w}_\infty = k_*^\top (K + \lambda I)^{-1} y$$



We show that

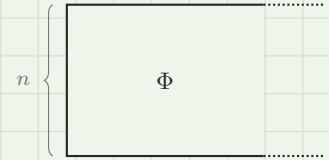
$$\varphi(x_*)^\top \hat{w}_\infty = k_*^\top (K + \lambda I)^{-1} y \approx \psi(x_*)^\top (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top y$$



Infinitely many columns

We show that

$$\varphi(x_*)^\top \hat{w}_\infty = k_*^\top (K + \lambda I)^{-1} y \approx \psi(x_*)^\top (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top y$$

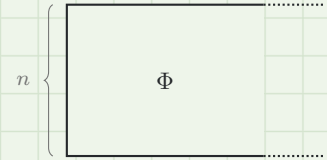




$$\Psi = \begin{bmatrix} — \ \psi(x_1)^\top \ — \\ \vdots \\ — \ \psi(x_n)^\top \ — \end{bmatrix} \in \mathbb{R}^{n \times 2Q}$$

We show that

$$\varphi(x_*)^\top \hat{w}_\infty = k_*^\top (K + \lambda I)^{-1} y \approx \psi(x_*)^\top (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top y$$



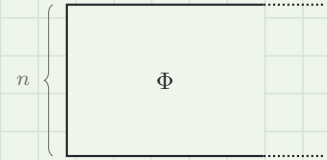


$$\Psi = \begin{bmatrix} — \ \psi(x_1)^\top \ — \\ \vdots \\ — \ \psi(x_n)^\top \ — \end{bmatrix} \in \mathbb{R}^{n \times 2Q}$$

and $2Q < n$

We show that

$$\varphi(x_*)^\top \hat{w}_\infty = k_*^\top (K + \lambda I)^{-1} y \approx \psi(x_*)^\top (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top y$$



Infinitely many columns



$2Q$

$$\Psi = \begin{bmatrix} — \ \psi(x_1)^\top \ — \\ \vdots \\ — \ \psi(x_n)^\top \ — \end{bmatrix} \in \mathbb{R}^{n \times 2Q}$$

and $2Q < n$

$\psi : \mathbb{R}^d \to \mathbb{R}^{2Q}$ is built in the next section

Fast polynomial OLSE

1) Compute $\psi(x_i)$ for each $i \leq n$

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1) Compute $\psi(x_i)$ for each $i \leq n$
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3) Compute $\Psi^\top \Psi + \lambda I$

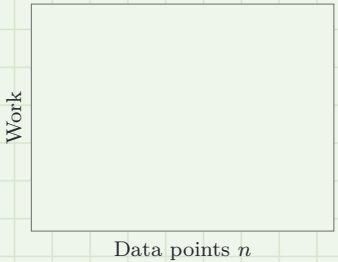
Fast polynomial OLSE

1) Compute $\psi(x_i)$ for each $i \leq n$
2) Given x_*, compute $\psi(x_*)$
3) Compute $\Psi^\top \Psi + \lambda I$
4) Compute $(\Psi^\top \Psi + \lambda I)^{-1}$

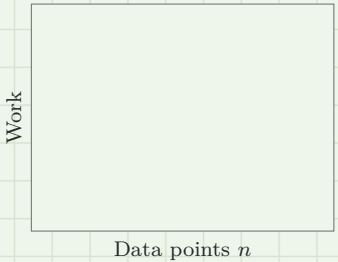
Fast polynomial OLSE

1) Compute $\psi(x_i)$ for each $i \leq n$
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4) Compute $(\Psi^\top \Psi + \lambda I)^{-1}$
5) Output $\psi(x_*)^\top (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top y$

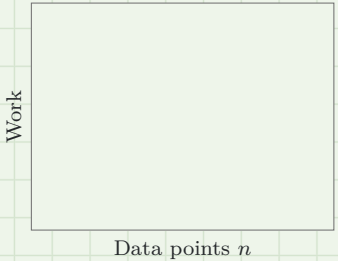
1. Compute every $\psi(x_i)$ $O(nQd)$



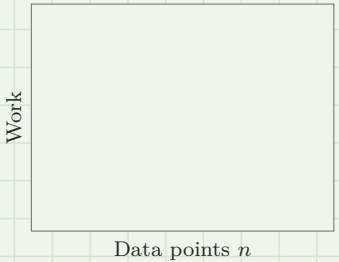
1. Compute every $\psi(x_i)$ $O(nQd)$
2. Compute $\psi(x_*)$ $O(Qd)$



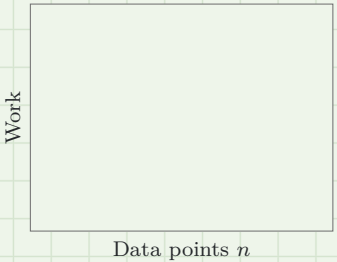
1. Compute every $\psi(x_i)$ $O(nQd)$
2. Compute $\psi(x_*)$ $O(Qd)$
3. Compute $\Psi^\top \Psi + \lambda I$ $O(nQ^2)$



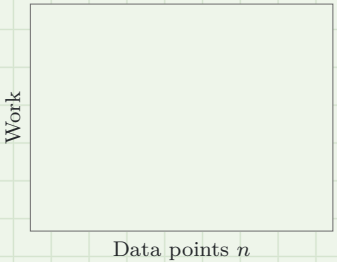
1. Compute every $\psi(x_i)$ $O(nQd)$
2. Compute $\psi(x_*)$ $O(Qd)$
3. Compute $\Psi^\top \Psi + \lambda I$ $O(nQ^2)$
4. Invert it $O(Q^3)$



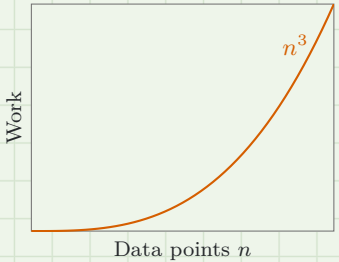
1. Compute every $\psi(x_i)$ $O(nQd)$
2. Compute $\psi(x_*)$ $O(Qd)$
3. Compute $\Psi^\top \Psi + \lambda I$ $O(nQ^2)$
4. Invert it $O(Q^3)$
5. Output $O(nQ + Q^2)$



1. Compute every $\psi(x_i)$	$O(nQd)$
2. Compute $\psi(x_*)$	$O(Qd)$
3. Compute $\Psi^\top \Psi + \lambda I$	$O(nQ^2)$
4. Invert it	$O(Q^3)$
5. Output	$O(nQ + Q^2)$
	$O(nQd + nQ^2 + Q^3)$

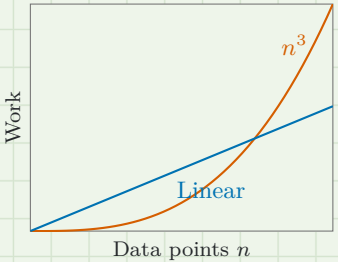


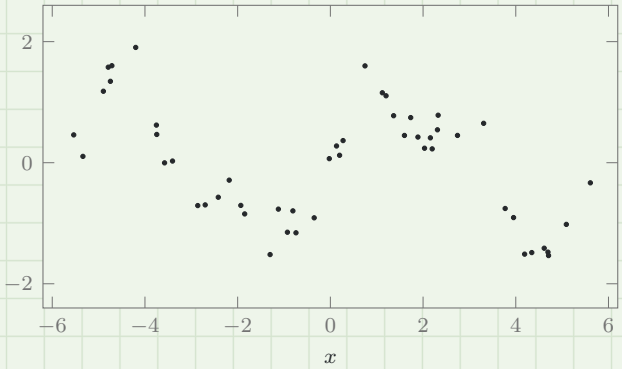
1. Compute every $\psi(x_i)$	$O(nQd)$
2. Compute $\psi(x_*)$	$O(Qd)$
3. Compute $\Psi^\top \Psi + \lambda I$	$O(nQ^2)$
4. Invert it	$O(Q^3)$
5. Output	$O(nQ + Q^2)$
	$O(nQd + nQ^2 + Q^3)$

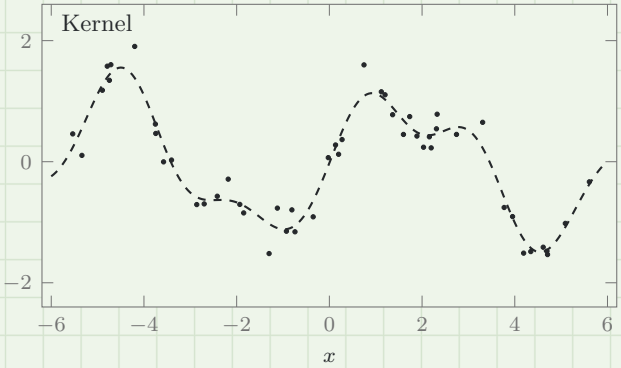


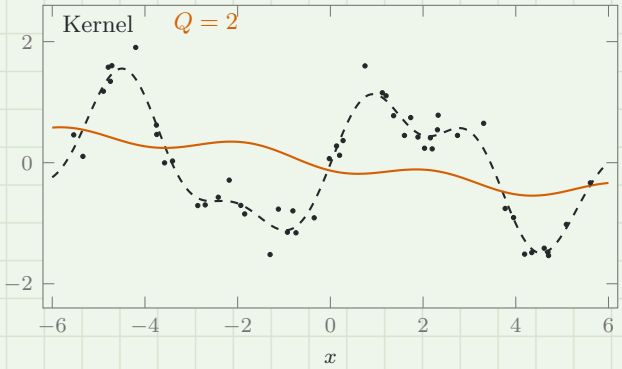
1. Compute every $\psi(x_i)$ $O(nQd)$
2. Compute $\psi(x_*)$ $O(Qd)$
3. Compute $\Psi^\top \Psi + \lambda I$ $O(nQ^2)$
4. Invert it $O(Q^3)$
5. Output $O(nQ + Q^2)$

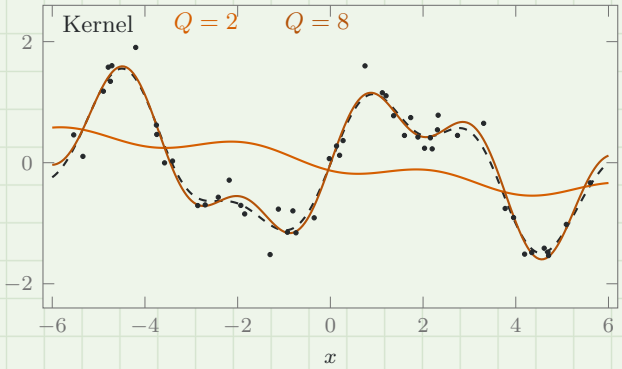
$$O(nQd + nQ^2 + Q^3)$$

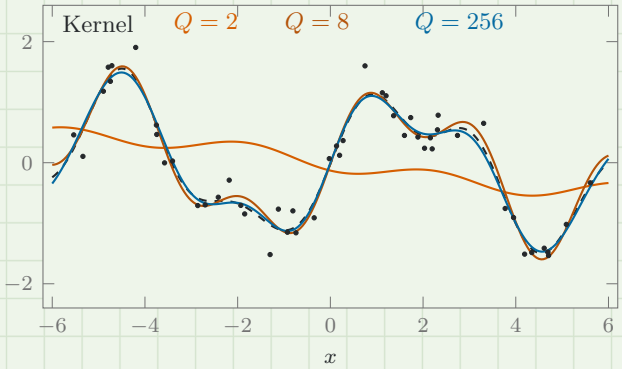



2
0
−2
−6
−4
−2
0
2
4
6
x


Kernel
2
0
−2
−6
−4
−2
0
2
4
6
x


Kernel
$Q = 2$
2
0
−2
−6
−4
−2
0
2
4
6
x


Kernel
$Q = 2$
$Q = 8$
2
0
−2
−6
−4
−2
0
2
4
6
x


Kernel
$Q = 2$
$Q = 8$
$Q = 256$
2
0
−2
−6
−4
−2
0
2
4
6
x

Entry 6

Why random features work

It suffices to show that $k(x, x') = \exp(-\frac{1}{2}\|x - x'\|^2) \approx \psi(x)^\top \psi(x')$

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Then $K \approx \Psi\Psi^\top$ and $k_* \approx \Psi\,\psi(x_*)$, so

$$\begin{aligned}&\varphi(x_*)^\top \Phi^\top (\Phi\Phi^\top + \lambda I)^{-1} y \\ &\quad \approx \psi(x_*)^\top \Psi^\top (\Psi\Psi^\top + \lambda I)^{-1} y\end{aligned}$$



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Then $K \approx \Psi\Psi^\top$ and $k_* \approx \Psi\,\psi(x_*)$, so

$$\begin{aligned}
&\varphi(x_*)^\top \Phi^\top (\Phi\Phi^\top + \lambda I)^{-1} y \\
&\quad \approx \psi(x_*)^\top \Psi^\top (\Psi\Psi^\top + \lambda I)^{-1} y \\
&\quad = \psi(x_*)^\top (\Psi^\top\Psi + \lambda I)^{-1} \Psi^\top y
\end{aligned}$$

$n \times n$

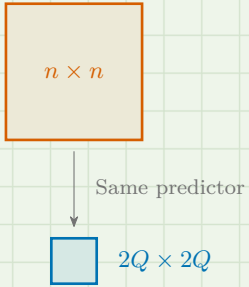
Push-through: $\Psi^\top(\Psi\Psi^\top + \lambda I) = (\Psi^\top\Psi + \lambda I)\Psi^\top$

It suffices to show that $k(x, x') = \exp(-\frac{1}{2}\|x - x'\|^2) \approx \psi(x)^\top \psi(x')$

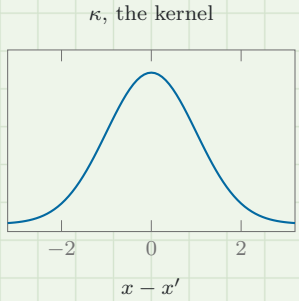
Then $K \approx \Psi\Psi^\top$ and $k_* \approx \Psi\,\psi(x_*)$, so

$$\begin{aligned}
&\varphi(x_*)^\top \Phi^\top (\Phi\Phi^\top + \lambda I)^{-1} y \\
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\end{aligned}$$

Push-through: $\Psi^\top(\Psi\Psi^\top + \lambda I) = (\Psi^\top\Psi + \lambda I)\Psi^\top$



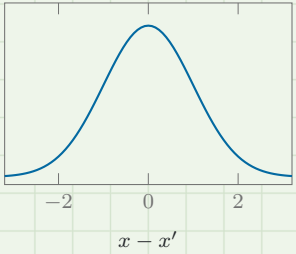
A continuous kernel that depends only on $x - x'$, with $k(x, x) = 1$, is the Fourier transform of a probability distribution



$$k(x, x') = \kappa(x - x') = \mathbb{E}_{\omega \sim p}\left[e^{\mathrm{i}\,\omega^\top (x - x')}\right]$$

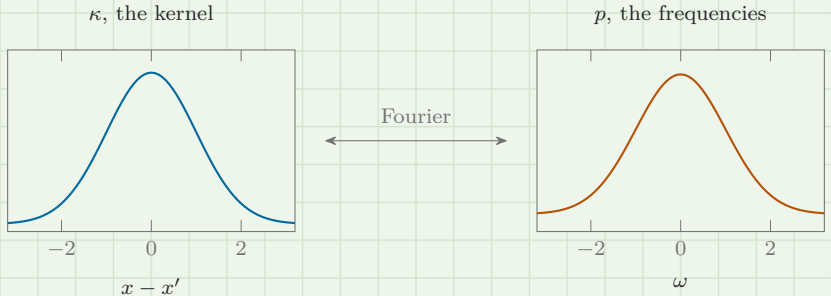
Bochner (1933); see Rudin, *Fourier Analysis on Groups* (1962). Used for features by Rahimi and Recht (2007)

κ, the kernel



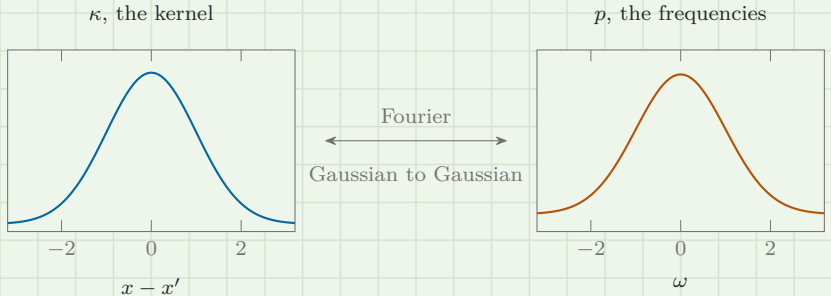
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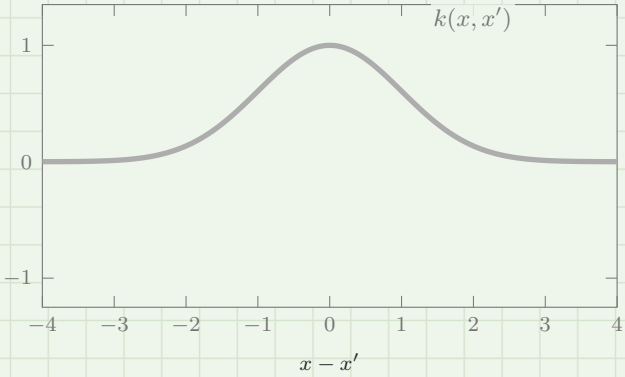
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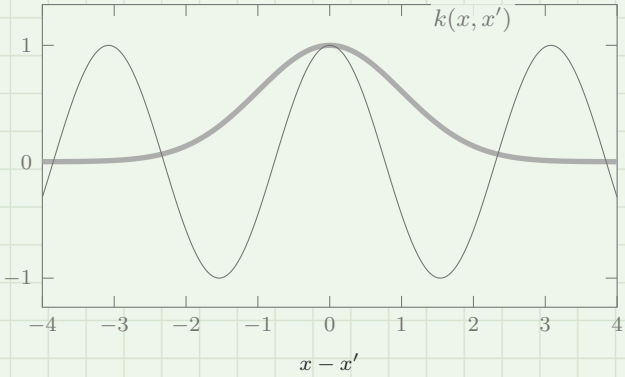


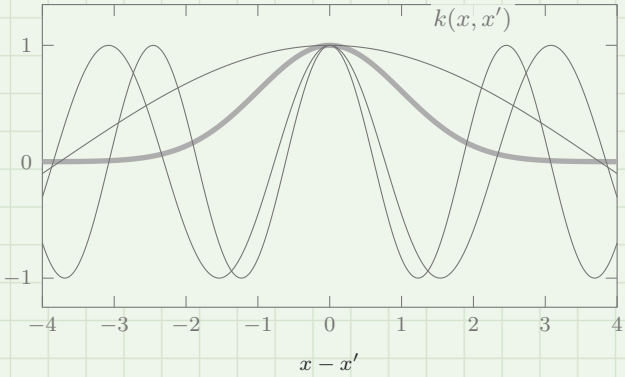
$$k(x, x') = \kappa(x - x') = \mathbb{E}_{\omega \sim p}\left[e^{\mathrm{i}\,\omega^\top (x - x')}\right]$$

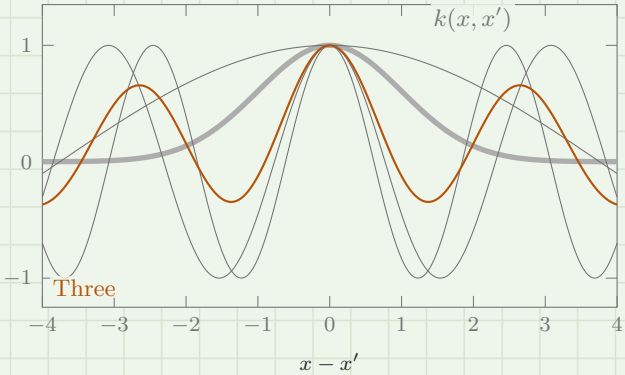
Bochner (1933); see Rudin, *Fourier Analysis on Groups* (1962). Used for features by Rahimi and Recht (2007)

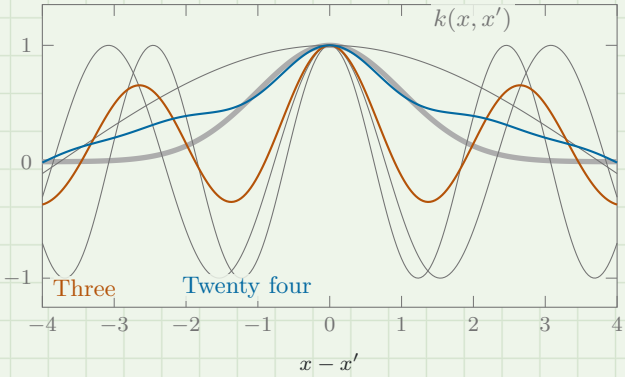


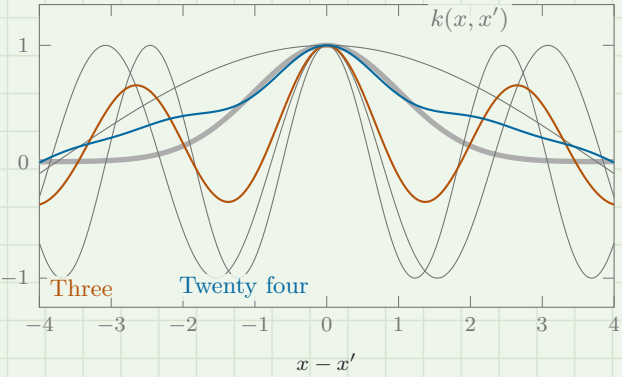

$k(x, x')$
1
0
−1
−4
−3
−2
−1
0
1
2
3
4
$x - x'$


$k(x, x')$
1
0
−1
−4
−3
−2
−1
0
1
2
3
4
$x - x'$


$k(x, x')$
1
0
−1
−4
−3
−2
−1
0
1
2
3
4
$x - x'$

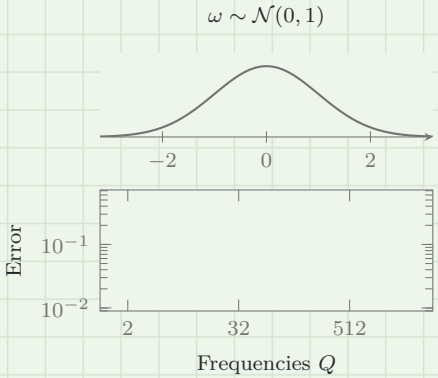

$k(x, x')$
1
0
−1
Three
−4
−3
−2
−1
0
1
2
3
4
$x - x'$


$k(x, x')$
1
0
−1
Three
Twenty four
−4
−3
−2
−1
0
1
2
3
4
$x - x'$



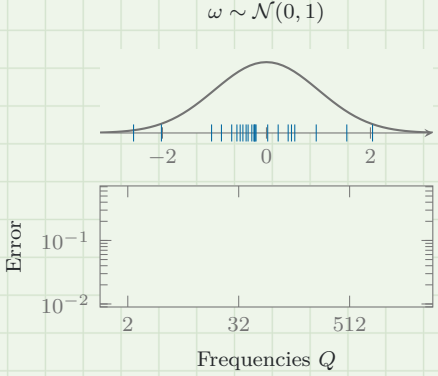
$$k(x, x') = \mathbb{E}_{\omega \sim \mathcal{N}(0, I_d)} \left[\cos\left(\omega^\top (x - x') \right) \right]$$

$$k(x, x') = \mathbb{E}_{\omega \sim \mathcal{N}(0, I_d)} \left[\cos\left(\omega^\top (x - x') \right) \right]$$



$$k(x, x') \approx \frac{1}{Q} \sum_{q \leq Q} \cos(\omega_q^\top (x - x'))$$

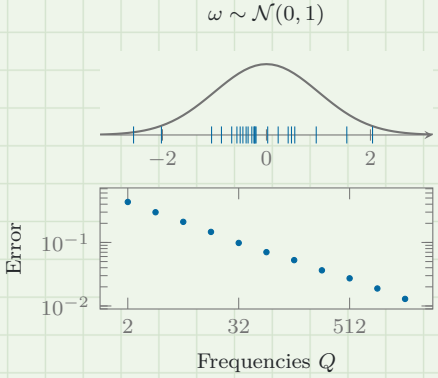
An average of Q independent draws, by the weak law of large numbers



- Drawn once, before any data

$$k(x, x') \approx \frac{1}{Q} \sum_{q \leq Q} \cos(\omega_q^\top (x - x'))$$

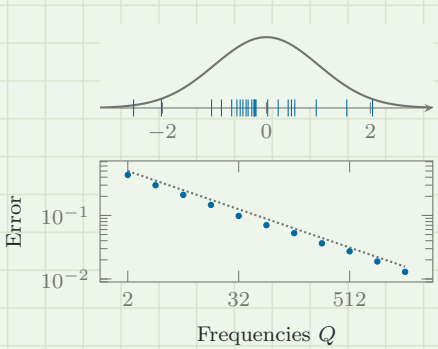
An average of Q independent draws, by the weak law of large numbers



- Drawn once, before any data
- Right on average

$$k(x, x') \approx \frac{1}{Q} \sum_{q \leq Q} \cos(\omega_q^\top (x - x'))$$

An average of Q independent draws, by the weak law of large numbers

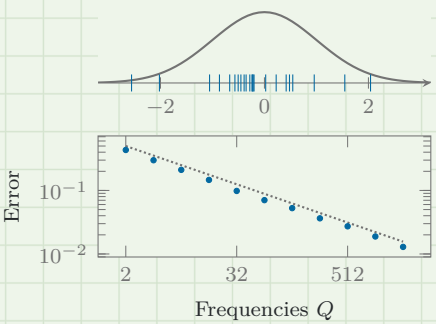


- Drawn once, before any data
- Right on average
- Error about $1/\sqrt{2Q}$

$$k(x, x') \approx \frac{1}{Q} \sum_{q \leq Q} \cos(\omega_q^\top (x - x'))$$

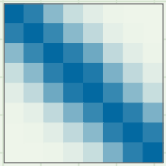
An average of Q independent draws, by the weak law of large numbers

$\omega \sim \mathcal{N}(0, 1)$

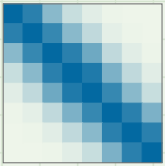


- Drawn once, before any data
- Right on average
- Error about $1/\sqrt{2Q}$
- $Q = 5000$: about 0.01

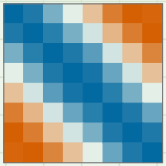
Kernel



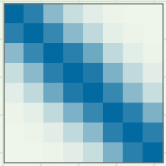
Kernel



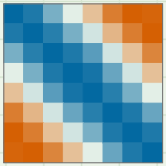
$Q = 1$



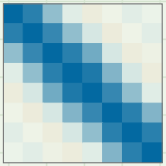
Kernel



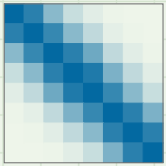
$Q = 1$



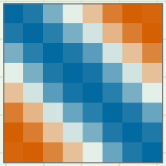
$Q = 5$



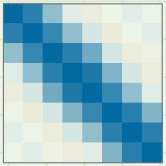
Kernel



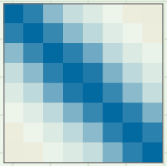
$Q = 1$



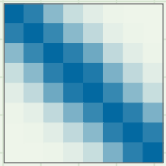
$Q = 5$



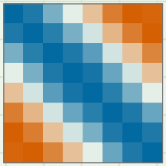
$Q = 50$



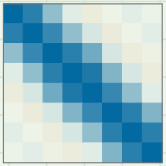
Kernel



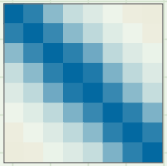
$Q = 1$



$Q = 5$

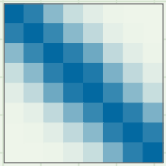


$Q = 50$

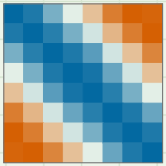


- One frequency: a single cosine, nothing like the kernel

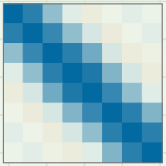
Kernel



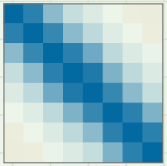
$Q = 1$



$Q = 5$



$Q = 50$



- One frequency: a single cosine, nothing like the kernel
- Fifty: the band structure is back

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

$$\cos(\omega^\top(x - x')) = \cos(\omega^\top x)\cos(\omega^\top x') + \sin(\omega^\top x)\sin(\omega^\top x')$$

$$\frac{1}{Q}\sum_{q\le Q}\cos\big(\omega_q^\top(x-x')\big) = \underbrace{\psi(x)^\top\psi(x')}_{\text{an inner product of } 2Q \text{ numbers}}$$

$$\frac{1}{Q}\sum_{q\le Q}\cos\big(\omega_q^\top(x-x')\big) = \underbrace{\psi(x)^\top\psi(x')}_{\text{an inner product of } 2Q \text{ numbers}}$$

$\cos(\omega_1^\top x)$
$\sin(\omega_1^\top x)$

$$\frac{1}{Q}\sum_{q\le Q}\cos\big(\omega_q^\top(x-x')\big) = \underbrace{\psi(x)^\top\psi(x')}_{\text{an inner product of } 2Q \text{ numbers}}$$

$\cos(\omega_1^\top x)$
$\sin(\omega_1^\top x)$
$\cos(\omega_2^\top x)$
$\sin(\omega_2^\top x)$
$\vdots$

$$\frac{1}{Q}\sum_{q\le Q}\cos\big(\omega_q^\top(x-x')\big) = \underbrace{\psi(x)^\top\psi(x')}_{\text{an inner product of } 2Q \text{ numbers}}$$

$$\psi(x) = \frac{1}{\sqrt{Q}}\begin{bmatrix}\cos(\omega_1^\top x)\\ \sin(\omega_1^\top x)\\ \cos(\omega_2^\top x)\\ \sin(\omega_2^\top x)\\ \vdots\end{bmatrix} \left.\vphantom{\begin{bmatrix}a\\b\\c\\d\\e\end{bmatrix}}\right\} 2Q \text{ entries}$$

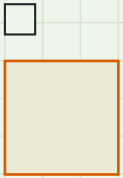
Entry 7

Summary

Linear $x \in \mathbb{R}^d$ $\square$

Linear $x \in \mathbb{R}^d$

Kernel $\varphi(x) \in \mathbb{R}^\infty$



Linear	$x \in \mathbb{R}^d$
Kernel	$\varphi(x) \in \mathbb{R}^\infty$
Random features	$\psi(x) \in \mathbb{R}^{2Q}$



Linear	$x \in \mathbb{R}^d$		$O(nd^2 + d^3)$
Kernel	$\varphi(x) \in \mathbb{R}^\infty$		
Random features	$\psi(x) \in \mathbb{R}^{2Q}$		

Linear	$x \in \mathbb{R}^d$		$O(nd^2 + d^3)$
Kernel	$\varphi(x) \in \mathbb{R}^\infty$		$O(n^2 d + n^3)$
Random features	$\psi(x) \in \mathbb{R}^{2Q}$		

Linear	$x \in \mathbb{R}^d$		$O(nd^2 + d^3)$
Kernel	$\varphi(x) \in \mathbb{R}^\infty$		$O(n^2 d + n^3)$
Random features	$\psi(x) \in \mathbb{R}^{2Q}$		$O(nQd + nQ^2 + Q^3)$

Linear	$x \in \mathbb{R}^d$		$O(nd^2 + d^3)$
Kernel	$\varphi(x) \in \mathbb{R}^\infty$		$O(n^2 d + n^3)$
Random features	$\psi(x) \in \mathbb{R}^{2Q}$		$O(nQd + nQ^2 + Q^3)$

- The kernel buys infinitely many features, at a cubic price

Linear	$x \in \mathbb{R}^d$		$O(nd^2 + d^3)$
Kernel	$\varphi(x) \in \mathbb{R}^\infty$		$O(n^2 d + n^3)$
Random features	$\psi(x) \in \mathbb{R}^{2Q}$		$O(nQd + nQ^2 + Q^3)$

- The kernel buys infinitely many features, at a cubic price
- Random features buy them back, at a linear one